

CS 2300 HW6

1) **Basis:** $P(1): 1^3 = 1$

$$(1 * (1 + 1)^2)/4 = 1$$
$$1=1$$

Thus, we can say that $P(1)$ is true.

Assume: $P(K) = 1^3 + 2^3 + \dots + k^3 = (k * (k + 1)/2)^2$

Prove: $P(K+1) = 1^3 + 2^3 + \dots + (K + 1)^3 = [(K + 1)(K + 2)/2]^2$

$$\begin{aligned} P(K+1) &= 1^3 + 2^3 + \dots + K^3 + (K + 1)^3 \\ &= P(K) + (K + 1)^3 \\ &= (K * (K + 1)/2)^2 + (K + 1)^3 \\ &= (K^2 * (K + 1)^2 / 4 + (K + 1)^3) \\ &= (K^2 * (K + 1)^2 + 4(K + 1)^3) / 4 \\ &= (K + 1)^2 * (K^2 + 4K + 4) / 4 \\ &= (K + 1)^2 * (K + 2)^2 / 4 \\ &= [(K + 1)(K + 2)/2]^2 \end{aligned}$$

Thus, since this statement holds true for $P(K+1)$, we can say that by mathematical induction, for all n , $P(n) = (n * (n + 1)/2)^2$.

2) **Basis:** $S(1) = 1*1! = 1$ (L.H.S)

$$2! - 1 \text{ (R.H.S)}$$

$$= 2 - 1$$

$$= 1$$

Thus, we can say that $S(1)$ is true, since both L.H.S and R.H.S are equal.

Assume: $P(K) = 1 * 1! + 2 * 2! + 3 * 3! + \dots + K * K! = (K + 1)! - 1$

Prove: $P(K + 1) = (K + 2)! - 1$

$$\begin{aligned} P(K + 1) &= 1 * 1! + 2 * 2! + 3 * 3! + \dots + K * K! + (K + 1) * (K + 1)! \\ &= (K + 1)! - 1 + (K + 1) * (K + 1)! \\ &= (K + 1)! + (K + 1)! * (K + 1) - 1 \\ &= (K + 1)! (1 + K + 1) - 1 \\ &= (K + 1)! (K + 2) - 1 \\ &= (K + 2)! - 1 \end{aligned}$$

Since, this statement holds true for $P(K+1)$, we can say that by mathematical induction, for all n , $P(n) = (K + 1)! - 1$

3) Basis: $P(0) = 2 * (-7)^0 = 2$ (L.H.S)
 $= (1 - (-7)^{0+1})/4$
 $= (1 + 7)/4$
 $= 8/4 = 2$ (R.H.S)

Thus, we can say that $P(0)$ is true, since L.H.S = R.H.S.

Assume: $P(K) = [1 - (-7)^{K+1}]/4$

Prove: $P(K + 1) = (1 - (-7)^{K+2})/4$

$$\begin{aligned} P(K + 1) &= 2 - 2 * 7 + 2 * (7)^2 - 2 * (7)^3 + \dots + 2(-7)^K + 2(-7)^{K+1} \\ &= [1 - (-7)^{K+1}]/4 + 2(-7)^{K+1} \\ &= [1 - (-7)^{K+1} + 8(-7)^{K+1}]/4 \\ &= [1 - (-7)^{K+1}(1 - 8)]/4 \\ &= [1 - (-7)^{K+1} * (-7)]/4 \\ &= [1 - (-7)^{K+2}]/4 \end{aligned}$$

Since, this statement holds true for $P(K+1)$, we can say that by mathematical induction for all n , $P(n) = (1 - (-7)^{n+1})/4$.

4) Basis: $P(1) = 1/1 * 2 = 1/2$ (L.H.S)
 $P(1) = 1/1+1 = 1/2$ (R.H.S)

Thus, we can say that this statement holds true for $P(1)$, since L.H.S is equal to R.H.S.

Assume: $P(K) = 1/1 * 2 + 1/2 * 3 + \dots + 1/K * (K + 1) = K/(K + 1)$

Prove:

$$P(K + 1) = 1/1 * 2 + 1/2 * 3 + \dots + 1/K(K + 1) + 1/(K + 1)(K + 2) = (K + 1)/(K + 1) + 1$$

$$\begin{aligned} P(K + 1) &= 1/1 * 2 + 1/2 * 3 + \dots + 1/K * (K + 1) + 1/[(K + 1) * (K + 2)] \\ &= K/(K + 1) + 1/[(K + 1) * (K + 2)] \\ &= (K^2 + 2K + 1)/[(K + 1) * (K + 2)] \\ &= (K + 1)^2/[(K + 1) * (K + 2)] \\ &= (K + 1)/(K + 2) \end{aligned}$$

$$= (K + 1) / [(K + 1) + 1]$$

Since we proved that this condition is true for $P(K+1)$, so we can say that, for all n ,
 $P(n) = 1/(1 * 2) + 1/(2 * 3) + \dots + 1/[n * (n + 1)] = n/[(n + 1)]$

5) Basis: $P(1) = 1+3+2 = 6$, which is divisible by 3, so we can say that $P(1)$ is true.

Assume: $P(K) = K^3 + 3K^2 + 2K$ is divisible by 3.

Prove: $P(K + 1) = (K + 1)^3 + 3 * (K + 1)^2 + 2 * (K + 1)$ is also divisible by 3.

$$\begin{aligned} P(K + 1) &= (K + 1)^3 + 3 * (K + 1)^2 + 2 * (K + 1) \\ &= K^3 + 1 + 3K + 3K^2 + 3 * (K^2 + 1 + 2K) + 2K + 2 \\ &= K^3 + 3K^2 + 2K + 3 * (K + 1)^2 + 3 \end{aligned}$$

Since we know that $K^3 + 3K^2 + 2K$ is divisible by 3, we can represent it as $3l$, for some integer l .

$$\begin{aligned} &= 3l + 3 * [(K + 1)^2 + 1] \\ &= 3(l + (K + 1)^2 + 1) \end{aligned}$$

Hence, we can say that $P(K + 1)$ is divisible by 3, so by mathematical induction we can say that for all n , $P(n)$ is divisible by 3.

6) Basis: $P(5) = 3^5 < 3(5!)$ (as 243 is less than 360, which we get after solving this equation.)

Assume: $P(K): 3^K < 3 * K!$

Prove: $P(K + 1): 3^{K+1} < 3[(K + 1)!]$

$$\begin{aligned} 3^{K+1} &= 3 * 3^K \\ 3^K &< 3 * K! \quad (\text{Our assumption}) \quad (1) \end{aligned}$$

Multiply both the sides of (1) by 3.

$$3 * 3^K < 3 * 3K!$$

We know that 3 is less than $K+1$, for $K > 5$

$$\begin{aligned}
3 * 3K! &< 3(K + 1)! \\
3 * 3K! &< 3(K + 1)K! \\
3 * 3K! &< (K + 1) * 3K!
\end{aligned}$$

Now, since $3 * 3^K < 3 * 3K!$ and $3 * 3K! < (K + 1) * 3K!$, so we can say that $3 * 3^K < (K + 1) * 3K!$

Thus, since this $P(K+1)$ is true, we can say that by mathematical induction, for all n , $P(n)$, $3^n < 3(n!)$.

7a) Basis: Let us take 3 base cases:

$$P(14) = 3*2 + 8*1 = 14$$

Since we were able to prove that 2 3-cent stamps and 1 8 cent stamp are needed to make 14 cents, we have proved that our basis is true.

Assume: Let us assume that this statement holds true for some integer K greater than equal to 14, such that it can be formed using 3-cent and 8-cent stamps.

Prove: $P(K+1)$ holds true

We have 2 cases:

- Some 8 cent stamps were used
- No 8 cent stamps were used

1st Case

Let us say that an 8 cent stamp was used in a K cent postage, so we can replace the 8 cent stamp with 3 3-cent stamps, which makes $k - 8 + (3*3) = k - 8 + 9 = k + 1$ cent postage.

2nd case

Let us assume that no 8-cent stamps were used, so that means only 3 cent stamps were used, so we can say that k is a multiple of 3. Since $K \geq 14$, and the next multiple of 3 that comes after 14 is 15, it follows that $k \geq 15$, so at least 5 3-cent stamps are used. In this case we must replace 5 3-cent stamps with 2 8-cent stamps so that we can form $K - (5*3) + (8*2) = K - 15 + 16 = K + 1$ cent postage.

Therefore, we can say that in the above 2 cases we have proved that $P(n)$ is true for all n greater than 14, by regular induction.

b) Basis step : $P(14) = (3*2) + (8*1)$
 $P(15) = (3*5) + (8*0)$
 $P(16) = (3*0) + (8*2)$

Over here, we can say that 14, 15 and 16 have unique usage of multiple of the 3-cent and 8-cent stamps, whereas 17, 18, 19 are not unique solutions, since they were made using 14, 15 and 16, so that's why we do not count them in our basis step.

Hence the terms that are used in our basis step are 14, 15 and 16.

Prove: $P(K+3)$ is true for all $K \geq 14$.

Since we have proved that by inductive step process, $P(K) \rightarrow P(K+3)$, we can say that $P(17) = P(14) + 3$, $P(18) = P(15) + 3$ and $P(19) = P(16) + 3$.

Hence we have proved that $P(n)$ is true for all $n \geq 14$, by strong induction.

8) We have been given 2 assumptions over here that are:

- i) $P(n)$ is true for an infinite number of positive integers
- ii) If $P(K+1)$ is true, then $P(K)$ is also true for all positive integers k .

Let us take a contrapositive of ii, so we have:

$$\neg P(K) \rightarrow \neg P(K+1)$$

Let us assume that there exists some integer m , such that $P(m)$ is false.

Contradiction Rule

Since $P(m)$ is false, $\neg P(m)$ is true and by proof of mathematical induction, $\neg P(n)$ is true, for every integer $n \geq m$. This contradicts our assumption that $P(n)$ is true for an infinite number of positive integers.

Therefore, we can say that $P(n)$ is true for all positive integers.